Introduction

Spatial transcriptomics assays (10x Visium, Slide-seq, Stereo-seq, MERFISH) measure gene expression with spatial coordinates. They bridge histology and scRNA-seq: the tissue architecture is preserved, so analyses like spatial domain detection, neighbourhood effects, and tumour-immune interface mapping become possible.

Prerequisites

scRNA-seq analysis basics; familiarity with tissue histology.

Theory

Visium spots contain ~10-100 cells each, giving spot-resolution (not single-cell) expression. Analysis proceeds like scRNA-seq but with two additions: (i) spatial variable-gene detection (Moran's I), and (ii) spatial domain inference (BayesSpace, Giotto, or spatial-aware clustering).

Deconvolution (cell2location, RCTD, SpatialExperiment) decomposes each spot into cell-type proportions using a matched scRNA-seq reference.

Assumptions

Spatial coordinates are correctly registered to H&E; spot-level expression is a linear combination of underlying cell types; tissue processing preserves mRNA integrity.

R Implementation

```r
library(Seurat)

set.seed(2026)
# Minimal spatial-like object: spots on a grid
n_spots <- 100
coords <- expand.grid(x = 1:10, y = 1:10)
counts <- matrix(rpois(200 * n_spots, lambda = 3), 200, n_spots)
rownames(counts) <- paste0("g", 1:200)
colnames(counts) <- paste0("spot_", 1:n_spots)

# Inject spatial pattern: top-left vs bottom-right
region <- ifelse(coords$x + coords$y < 10, "A", "B")
counts[1:20, region == "A"] <- counts[1:20, region == "A"] + 8

so <- CreateSeuratObject(counts)
so$x <- coords$x; so$y <- coords$y; so$region <- region

so <- NormalizeData(so, verbose = FALSE)
so <- FindVariableFeatures(so, nfeatures = 100, verbose = FALSE)
so <- ScaleData(so, verbose = FALSE)
so <- RunPCA(so, npcs = 10, verbose = FALSE)
so <- FindNeighbors(so, dims = 1:10, verbose = FALSE)
so <- FindClusters(so, resolution = 0.5, verbose = FALSE)

plot(coords, col = factor(Idents(so)), pch = 15, cex = 2,
     main = "Spot clusters on tissue coordinates",
     xlab = "x", ylab = "y")
```

Output & Results

A clustered spatial object; the 2-D scatter shows clusters aligning with spatial regions, recapitulating tissue architecture.

Interpretation

"Spatial clustering of the Visium section identified six domains corresponding to tumour core, tumour edge, stroma, and three immune niches; interface enrichment analysis revealed T-cell accumulation at the tumour-stroma boundary."

Practical Tips

- Always overlay clusters on H&E to validate biological interpretation.
- Moran's I (`SpatialExperiment::spatialCorrelation` or `Seurat::FindSpatiallyVariableFeatures`) finds spatially variable genes.
- Deconvolution is essential when Visium spots cover multiple cell types; use a matched scRNA-seq reference.
- BayesSpace and SpaGCN integrate spatial structure into clustering, giving smoother domain boundaries.
- Stereo-seq and MERFISH offer sub-cellular resolution; analysis pipelines differ slightly from spot-based Visium.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat